Mario's Diet Problem Solver

Mario's Diet Problem Solver is a Shiny-based web application designed to optimize food selection for a cost-effective and nutritionally balanced diet. Using the Simplex algorithm, the application identifies the optimal servings of selected foods to meet predefined nutritional constraints while minimizing costs.

Application Structure

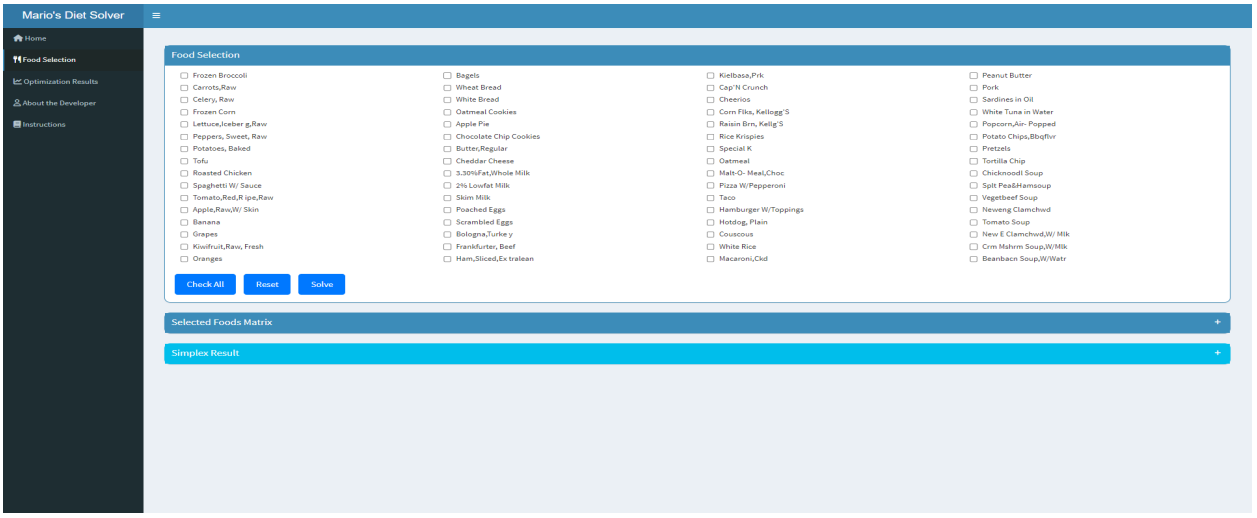
The app is divided into several sections:

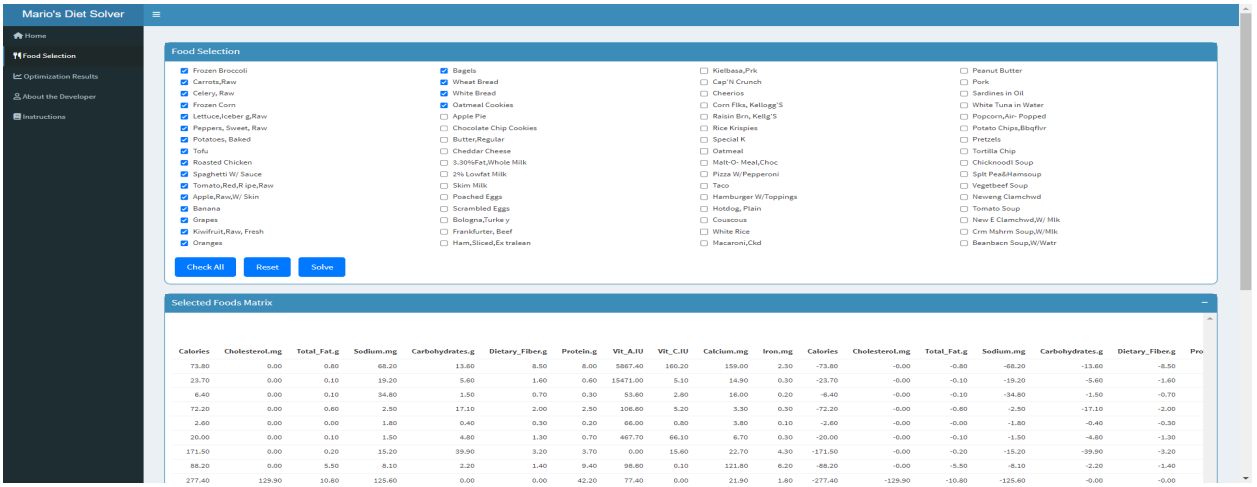
1. Home Page

- Displays a welcoming interface with an image of Mario.
- Provides an introduction to the application.

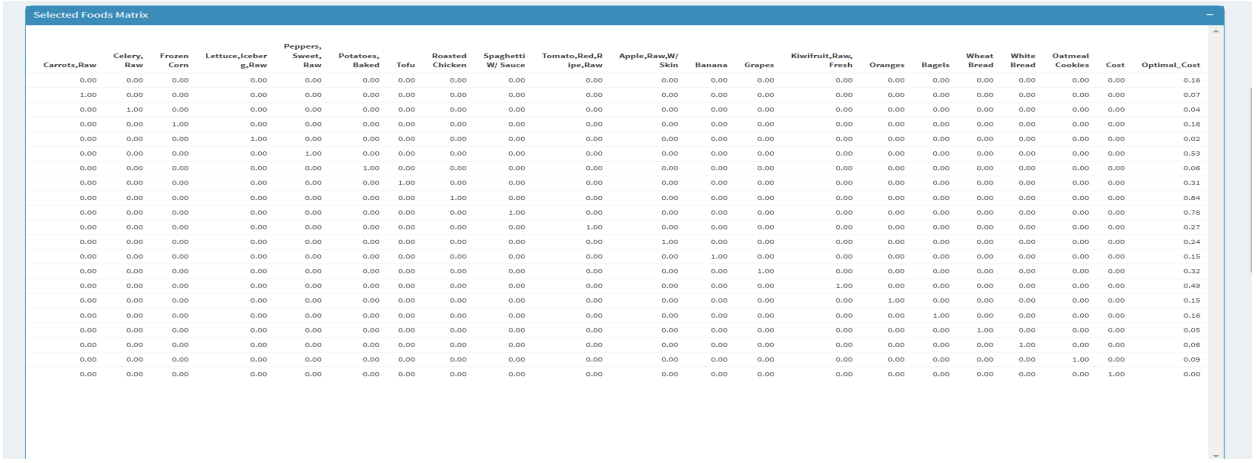
2. Food Selection

- Allows users to select foods from the nutritional database.
- Options include:
  - **Check All**: Select all food items.
  - **Reset**: Clear all selected items.
  - **Solve**: Optimize the selected foods using the Simplex algorithm.
- Displays:
  - **Selected Foods Matrix**: Shows the initial tableau before passing it down to the Simplex function.
  - **Simplex Result**: Displays the final Simplex tableau.





Calories	Cholesterol.mg	Total_Fat.g	Sodium.mg	Carbohydrates.g	Dietary_Fiber.g	Protein.g	Vit_A.IU	Vit_C.IU	Calcium.mg	Iron.mg	Calories	Cholesterol.mg	Total_Fat.g	Sodium.mg	Carbohydrates.g	Dietary_Fiber.g	Pro
73.80	0.00	0.80	68.20	13.60	8.50	8.00	5867.40	160.20	159.00	2.30	-73.80	-0.00	-0.80	-68.20	-13.60	-8.50	
23.70	0.00	0.10	19.20	5.60	1.60	0.60	15471.00	5.10	14.90	0.30	-23.70	-0.00	-0.10	-19.20	-5.60	-1.60	
6.40	0.00	0.10	34.80	1.50	0.70	0.30	53.60	2.80	16.00	0.20	-6.40	-0.00	-0.10	-34.80	-1.50	-0.70	
72.20	0.00	0.60	2.50	17.10	2.00	2.50	106.60	5.20	3.30	0.30	-72.20	-0.00	-0.60	-2.50	-17.10	-2.00	
2.60	0.00	0.00	1.80	0.40	0.30	0.20	66.00	0.80	3.80	0.10	-2.60	-0.00	-0.00	-1.80	-0.40	-0.30	
20.00	0.00	0.10	1.50	4.80	1.30	0.70	467.70	66.10	6.70	0.30	-20.00	-0.00	-0.10	-1.50	-4.80	-1.30	
171.50	0.00	0.20	15.20	39.90	3.20	3.70	0.00	15.60	22.70	4.30	-171.50	-0.00	-0.20	-15.20	-39.90	-3.20	
88.20	0.00	5.50	8.10	2.20	1.40	9.40	98.60	0.10	121.80	6.20	-88.20	-0.00	-5.50	-8.10	-2.20	-1.40	
277.40	129.90	10.80	125.60	0.00	0.00	42.20	77.40	0.00	21.90	1.80	-277.40	-129.90	-10.80	-125.60	-0.00	-0.00	



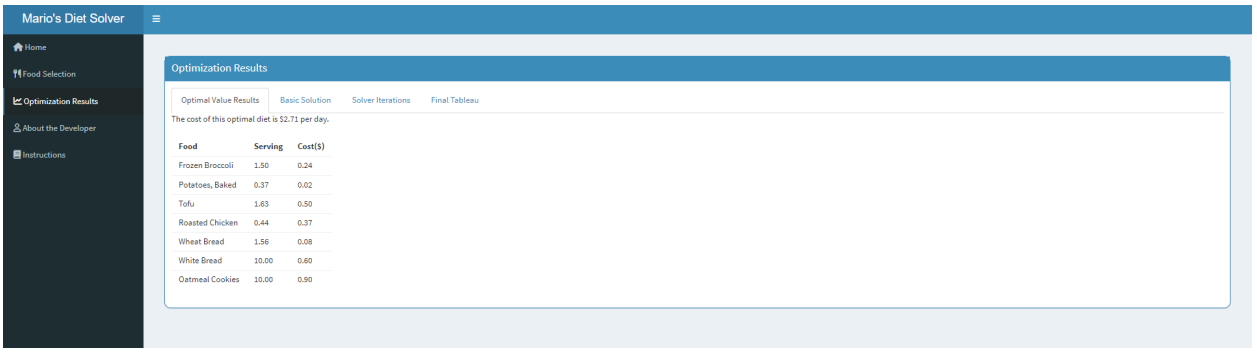
Carrots,Raw	Celery, Raw	Frozen Corn	Lettuce,Iceber g,Raw	Peppers, Sweet, Raw	Potatoes, Baked	Tofu	Roasted Chicken	Spaghetti W/ Sauce	Tomato,Red,R ipe,Raw	Apple,Raw,W/ Skin	Banana	Grapes	Kiwifruit,Raw, Fresh	Oranges	Bagels	Wheat Bread	White Bread	Oatmeal Cookies	Cost	Optimal_Cost
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.16
1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.07
0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.04
0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.18
0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.02
0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.53
0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.06
0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.31
0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.84
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.78
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.27
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.24
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.15
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.32
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.49
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.15
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.16
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.05
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.06
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.09
0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00

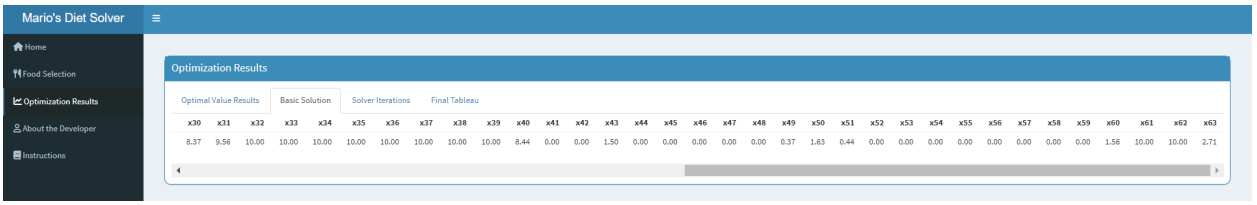
Simplex Result

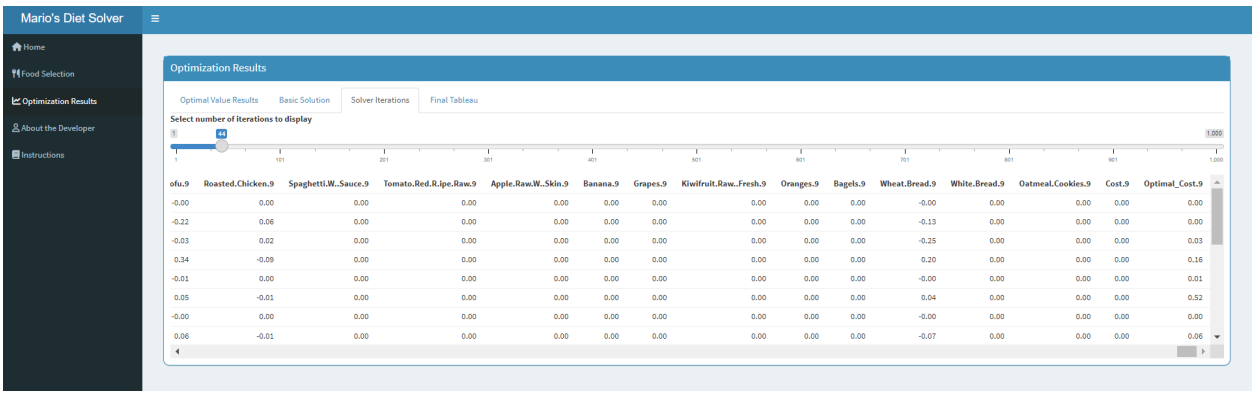
Carrots,Raw	Celery, Raw	Frozen Corn	Lettuce,Iceberg,Raw	Peppers, Sweet, Raw	Potatoes, Baked	Tofu	Roasted Chicken	Spaghetti W/ Sauce	Tomato,Red,Ripe,Raw	Apple,Raw,W/ Skin	Banana	Grapes	Kiwifruit,Raw, Fresh	Oranges	Bagels	Wheat Bread	White Bread	Oatmeal Cookies	Cost	Optimal_Cost
0.00	0.00	0.00	0.00	0.00	-0.00	-0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-0.00	0.00	0.00	0.00	0.00
0.00	0.00	0.00	0.00	0.00	-0.01	-0.22	0.06	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-0.13	0.00	0.00	0.00	0.00
0.00	1.00	0.00	0.00	0.00	0.07	-0.03	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-0.25	0.00	0.00	0.00	0.03
0.00	0.00	1.00	0.00	0.00	-0.44	0.34	-0.09	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.20	0.00	0.00	0.00	0.16
0.00	0.00	0.00	1.00	0.00	-0.00	-0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-0.00	0.00	0.00	0.00	0.01
0.00	0.00	0.00	0.00	1.00	-0.11	0.05	-0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.04	0.00	0.00	0.00	0.52
0.00	0.00	0.00	0.00	0.00	0.00	-0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-0.00	0.00	0.00	0.00	0.00
1.00	0.00	0.00	0.00	0.00	-0.08	0.06	-0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-0.07	0.00	0.00	0.00	0.06
0.00	0.00	0.00	0.00	0.00	0.18	-0.16	0.12	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.34	0.00	-1.00	0.00	0.01
0.00	0.00	0.00	0.00	0.00	1.45	-0.43	0.12	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-9.58	0.00	0.00	0.00	0.40
0.00	0.00	0.00	0.00	0.00	-0.12	-0.00	-0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	-0.06	0.00	0.00	0.00	0.26
0.00	0.00	0.00	0.00	0.00	-0.53	0.45	-0.09	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.29	0.00	0.00	0.00	0.23
0.00	0.00	0.00	0.00	0.00	-0.68	0.53	-0.11	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.32	0.00	0.00	0.00	0.14
0.00	0.00	0.00	0.00	0.00	-0.10	0.07	-0.01	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.05	0.00	0.00	0.00	0.32
0.00	0.00	0.00	0.00	0.00	-0.26	0.22	-0.05	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.17	0.00	0.00	0.00	0.47
0.00	0.00	0.00	0.00	0.00	-0.32	0.42	-0.08	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.40	0.00	0.00	0.00	0.11
0.00	0.00	0.00	0.00	0.00	-0.02	-0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	-1.10	0.00	0.00	0.00	0.10
0.00	0.00	0.00	0.00	0.00	-0.03	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.93	-1.00	0.00	0.00	0.01
0.00	0.00	0.00	0.00	0.00	-0.02	0.02	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-0.00	0.00	0.00	0.00	0.01
0.00	0.00	0.00	0.00	0.00	0.00	-0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-0.01	0.00	0.00	0.00	0.00
0.00	0.00	0.00	0.00	0.00	0.37	1.63	0.44	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.56	10.00	10.00	1.00	2.71

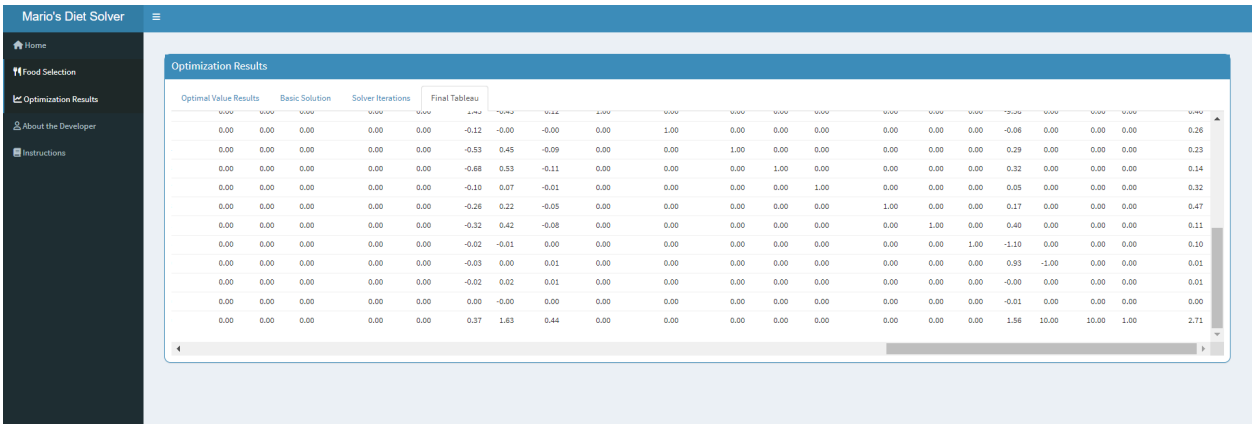
3. Optimization Results

- Summarizes the optimization process with:
  - **Optimal Value Results**: Displays the minimum cost and servings per food.
  - **Basic Solution**: Shows the solution for the selected variables.
  - **Solver Iterations**: Presents intermediary tableaux for each iteration.
  - **Final Tableau**: Shows the concluding Simplex tableau.







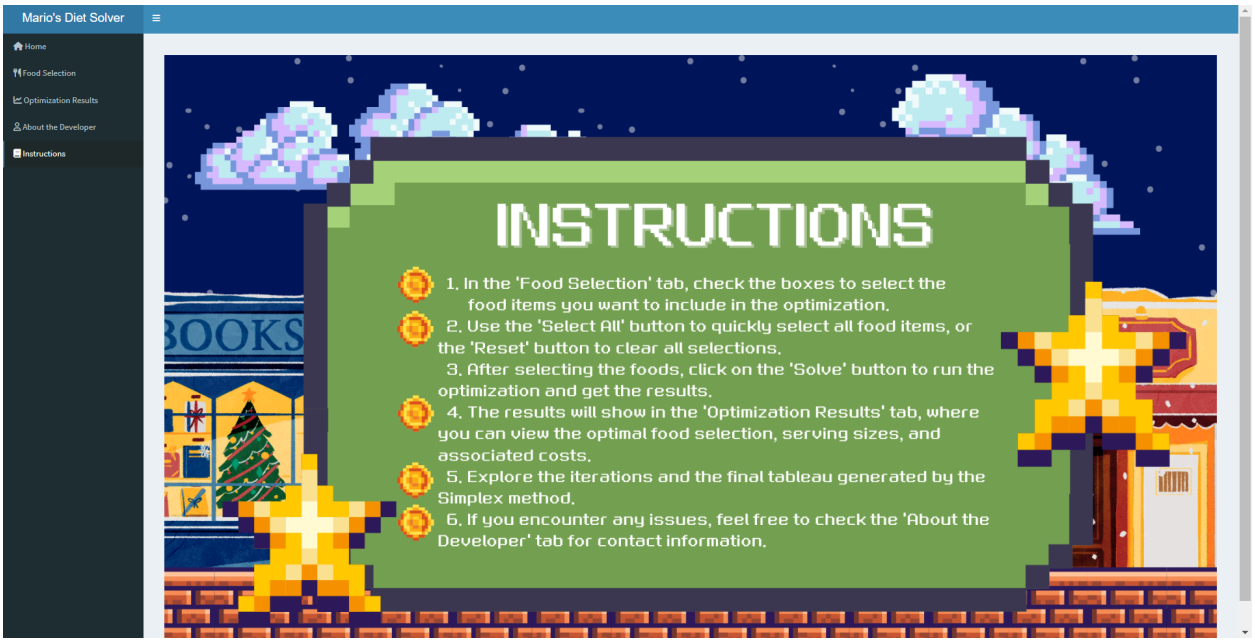


4. About the Developer

- Features a short description or image about the developer.

5. Instructions

- Provides guidance on how to use the app.



How to Use the Application

Step 1: Launch the Application

- Open the Shiny app either locally in RStudio or on a web server hosting the app.

Step 2: Navigate to Food Selection

- Click on the "Food Selection" tab in the sidebar menu.

Step 3: Select Foods

- Choose food items using the **checkboxes** under the food categories.
- Use action buttons for faster operations:
  - **Check All**: Select all foods.
  - **Reset**: Clear all selections.

Step 4: Solve the Optimization

- After selecting foods, click **Solve** to compute the optimal servings.
- Wait for the application to process the selected foods using the Simplex algorithm.

Step 5: Review Results

- Navigate to the **Optimization Results** tab.

- Explore the following:
    - **Optimal Value Results**: Check cost and serving breakdown.
    - **Basic Solution**: Review the selected variables and their values.
    - **Solver Iterations**: Use the slider to view intermediary Simplex steps.
    - **Final Tableau**: Analyze the final tableau of the Simplex algorithm.

Troubleshooting

1. **No Foods Displayed in Matrix**

    - Ensure at least one food item is selected.
    - Use the "Check All" button to select all items if unsure.
2. **Error in Optimization**

    - Ensure the food list is not empty when clicking **Solve**.
    - Check that your constraints (e.g., nutritional requirements) are realistic.
3. **Table or Graph Not Displaying Properly**

    - Reload the application or clear the browser cache.
4. **No Feasible Solution**

    - Adjust the selection of foods to ensure the constraints can be satisfied.

Additional Features

- **Scroll and Collapsible Boxes**: All large data outputs are scrollable for better usability.
- **Interactive Tabs**: Easily navigate between tabs for different functionalities.
- **Iteration Slider**: Control how many iterations are displayed during the optimization process.

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